

SUPPLEMENTARY MATERIAL

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Supplementary Material: Agentic Autoresearch for CT Reconstruction

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Received: 9 July 2026 | **Revised:** 9 July 2026 | **Accepted:** 9 July 2026

Keywords: CT reconstruction | sparse-view | low-dose | known operators | benchmarking | noise robustness

ABSTRACT

This document contains the supplementary material for *Agentic Autoresearch for CT Reconstruction*. It provides the full solver taxonomy and per-method configuration (S1), the Mayo-LDCT data, split, and geometry pipeline (S2), the breast split and train/test leakage audit (S3), the Mayo test board and significance (S4), the parameter-efficient recombination study (S5), the full breast noiseless and noisy boards (S6), an extended account of agent behavior (S7), and the list of additional supplementary figures (S8).

TABLE S1 | Full solver taxonomy. Axis A = physics engagement; Axis B = prior source. The final column cites the source(s) for each family. Abbreviations: DC, data consistency; N2I, Noise2Inverse; rectified-flow prior, a fast diffusion-type generative prior; FoE, field of experts (a learned filter-bank image prior).

Family	Representative solver(s)	Data-consistency D	Prior $\mathcal{R}$	Axis A	Axis B	Ref.
Classical iterative	tv-iterative, manhart-PWLS-TV	filtered / SART grad	total variation	in-loop	hand-crafted	1,2
Image-domain denoiser	manduca-bilateral, DD-UNet sup.	none (post-FBP)	bilateral / U-Net	none	hand-cr. / sup.	3,4
Unrolled learned-iterative	ITNet v1/v2/v3, U-Swin	in-loop DC	learned CNN / transformer	in-loop	supervised	5,6,7,8,9
Variational network	HammerNet-2017, HammerNet-VN	in-loop DC	field-of-experts	in-loop	supervised	10,11
Learned primal-dual	learned-primal-dual (LPD)	filtered + learned dual	learned combination	in-loop	supervised	12
Dual-domain	DD-supervised, DD-bilateral, DD-N2I	image + sinogram	sup. / bilateral / self-sup.	single/in-loop	sup. / self-sup.	13,14
Generative prior	fastdiff-flow/navisnet v4	DC + diffusion	rectified-flow prior	in-loop	generative	15,16,17,18
Per-scene implicit	NAF, R ² -Gaussian	per-scene fit	implicit representation	per-scene	implicit	19,20,21
Foundation zero-shot	RAM	single DC	frozen foundation model	single	foundation	22
Band decomposition	Wu-2015, Wu-2015-trainable	band-weighted DC	frequency-band residual	in-loop	hand-cr. / sup.	2
Recombination (ours)	param-efficient	filtered + LPD combine	bilateral post-filter	in-loop	hybrid	—

S1 | Solver taxonomy and per-method configuration (Table 1, full)

We place the 26 methods on two axes. Axis A is physics engagement (how the operator **A** is used at inference). Axis B is the prior source $\mathcal{R}$.

Full per-solver CONFIG defaults, parameter counts, and design notes are in each solver's design doc under `pentathlon/demo_dl_reference/` and the observation records under `docs/runs/`.

S2 | Mayo-LDCT — data, split, and the geometry pipeline

Real helical low-dose CT from the 2016 AAPM Low-Dose CT Grand Challenge.²³ We rebinned helical to fan-beam by single-slice rebinning.²⁴ The differentiable fan-beam projector used at every step is PYRO-NN.²⁵ This pipeline was the dominant human cost of the whole study (~24 active working days; three travel gaps excluded). We use a fixed patient split: train L145/186/209/219, validation L277, test L014/056/058/075/123. Test scores are the per-patient mean $\pm$ std over the five test patients ($n = 5$). Geometry details, rebinning validation, and the effort timeline figure are in the repository (`docs/runs/mayo_effort_timeline.png`). The Mayo challenge is one of a family of AAPM reconstruction grand challenges spanning sparse-view, spectral, truth-based, and metal-artifact tasks.^{26,27,28,29,30}

S3 | Breast split and the train/test leakage audit

Split. We re-partitioned the public 4000-case Sidky train pool into train (3600), validation (200), and test (200). The split is deterministic (seed 20260703) and disjoint. The official challenge val/test truths are withheld, so we cannot use them; our test split is drawn from the public pool.

Leakage audit (three independent checks).

TABLE S2 | Breast versus Mayo: the two problems are limited by different things, so their absolute hr scales are not comparable.

	Breast (Sidky)	Mayo
data	noiseless, ideal	real low-dose, quantum noise
bottleneck	data incompleteness	noise / dose
RMSE floor	0	> 0
best hr	~0.89	~0.38
dominant lever	data-consistency / known operator	denoising

TABLE S3 | Parameter-efficient recombination arc on breast (128-view sparse), all under the 20-min budget.

step	architecture	params	val hr
image-domain unroll (any config)	Mayo-style FoE denoiser	~1930	≤ 0.26 (ceiling)
filtered DC FBP(Ax - g)	K = 30	13	0.4345
+ learned primal-dual combination	W = 4	183	0.5047
+ 3-scale bilateral output filter	$\sigma_r = 0.02$	195	0.6201 (single best)

1. *Mechanism.* Training loads the train split. The test redirect fires only for the evaluation load. Training never sees test.
2. *Pixel-hash disjointness.* Over all 3600 train, 200 val, and 200 test truth images: $\text{TEST} \cap \text{TRAIN} = \emptyset$, $\text{TEST} \cap \text{VAL} = \emptyset$, $\text{VAL} \cap \text{TRAIN} = \emptyset$.
3. *Behavioral.* Test-hr tracks val-hr for every method, within ± 0.01 , and several methods score *lower* on test than val. Training on test would make test-hr much higher. It does not. The high absolute hr is the dataset (noiseless, well-posed), not a leak.

Why breast hr is high, and not comparable to Mayo. The Sidky data is noiseless by design; its RMSE floor is zero. Breast is incompleteness-limited (128 of ~1000 views). Mayo is noise-limited (real quantum noise). The two absolute hr scales are therefore not comparable.

S4 | Mayo test board and significance

Champion ITNet v1, test hr 0.376 ($n = 5$). The top tier is a statistical tie: ITNet v1 / ITNet v2 / U-Swin at the 5% level; the compact param-efficient solver joins at the 1% level ($p = 0.02$ – 0.027). At $n = 5$ the paired t -test has low power (a Wilcoxon test cannot even reach $p < 0.05$). Full per-solver table, per-patient hr, and the pairwise significance matrix are in docs/runs/mayo_significance_stats.md and the forest/matrix figures.

S5 | The parameter-efficient recombination study (search arc)

The agent recombined the method parts into a minimal-parameter solver. The climb was made of genuine six-box iterations. Key steps on breast (128-view sparse), all under the 20-min budget:

Findings, each earned by an iteration: the image-domain denoiser caps at ~0.26; deeper *adjoint* data-consistency lowers hr (it re-injects streaks); the *filtered* gradient breaks the ceiling at 13 parameters; an in-loop prox (proximal / denoising step) hurts (the filtered gradient already reconstructs); a learned cross-step combination helps; a full conjugate-gradient DC solve does not beat

TABLE S4 | Breast noiseless test board ($n = 200$).

#	solver	test hr	$\pm$ std	best iter
1	dual-domain-supervised	0.8948	0.0127	20
2	itnet	0.8926	0.0150	16
3	itnet-v2	0.8893	0.0151	15
4	itnet-v3	0.8749	0.0157	12
5	uswin	0.8586	0.0153	20
6	learned-primal-dual	0.7233	0.0143	12
7	hammernik-2017	0.6265	0.0135	12
8	param-efficient (195 p)	0.6212	0.0076	28
9	hammernik-vn	0.5787	0.0119	16
10	fastdiff-flow-pixel-constrained	0.5119	0.0211	15
11–23	ram · tv-iterative · manhart-PWLS-TV · wu-2015-trainable · manduca-bilateral · fastdiff* · dd-bilateral-sup · wu-2015 · tv-iter-sup · dd-bilateral-n2i · dd-n2i: hr 0.37 → 0.00			
DNF	naf, r2gaussian (per-scene; ~14–20/200 cases under budget)			

the single filtered gradient under a fixed wall; and Wu-2015 band machinery adds nothing here. The champion is seed-fragile in the variational-network manner: 4 of 5 seeds cluster at 0.571 ± 0.033 , one collapses; two stabilizers did not recover it. On the held-out test set the compact solver scores 0.6212 ± 0.0076 — the tightest per-case std of any solver.

Mayo vs breast contrast. On Mayo the compact optimum is an evolved field-of-experts plus multi-scale bilateral denoiser (hr 0.324 at ~0.00097 M params), statistically tied with the top tier at the 1% level. On a denoising problem, a tiny denoiser suffices. On sparse-view breast, that same denoiser fails; data-consistency is the differentiator.¹² This problem-dependence is the practical face of the known-operator principle^{31,32}: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it curbs the hallucination risk of unconstrained learned maps.^{33,34} The agent re-derived each optimum; it did not transfer.

Human role in this study. The recombination *strategy* was a human idea, on both datasets. On Mayo the human repeatedly asked whether more parameter-efficient approaches existed and explicitly suggested combining the pieces. On breast the finale was framed as a cross-method recombination and steered away from copying Mayo. The agent executed and mechanistically refined; it did not originate the recombination strategy.

S6 | Breast boards — noiseless and noisy (full)

S6.1 | Noiseless test board ($n = 200$, hr mean $\pm$ std)

S6.2 | Noisy test board ($I_0 = 10^5$, same models, no retraining)

Reading the reversal by framework axes. Robust: in-loop physics (learned-primal-dual) or hand-crafted smoothing priors (bilateral, total variation). Brittle: supervised image-domain maps trained on clean FBP (dual-domain-supervised; the constrained fastdiff variants). The hr metric is fair *within* the noisy board — all methods share the same noisy FBP baseline. The absolute SSIM/PSNR columns confirm the ranking is genuine, not baseline-gaming.

TABLE S5 | Breast noisy test board ($I_0 = 10^5$, same trained models, no retraining).

#	solver	noisy hr	noisy SSIM	nl. hr	chg.
1	learned-primal-dual	0.9349	0.9905	0.7233	↑ (6→1)
2	manduca-bilateral	0.8420	0.9537	0.276	↑↑
3	dual-domain-bilateral-n2i	0.7927	0.9532	0.206	↑↑
4	hammernik-2017	0.7016	0.7178	0.6265	↑
5	itnet-v3	0.7007	0.7564	0.8749	↓
6	fastdiff-flow-pixel-uncon.	0.6620	0.7058	0.269	↑
7	ram-zeroshot	0.6382	0.6578	0.295	↑
8	uswin	0.5915	0.6408	0.8586	↓
9	itnet	0.5784	0.6295	0.8926	↓
10	hammernik-vn	0.5529	0.5962	0.5787	≈
11	itnet-v2	0.5480	0.6201	0.8893	↓
12	param-efficient (195 p)	0.5153	0.5771	0.6212	↓ (holds)
13–14	manhart-PWLS-TV, tv-iterative:	hr 0.5068, SSIM 0.6164	(↑)		
15–18	wu-2015, wu-2015-trainable, tv-iter-sup, dd-bilateral-sup:	0.37 → 0.10			
low	dual-domain-supervised:	hr 0.0000 , SSIM 0.3483, nl. 0.8948	(↓↓, 1→last)		
low	fastdiff-wdm-* constr., fastdiff-flow-pixel-constr.:	0.00–0.64 (some incomplete)			
DNF	naif, r2gaussian (per-scene, incomplete)				

S6.3 | Noise model and no-retrain protocol

Poisson photon noise per Eq. (6) of the main text, $I_0 = 10^5$, applied to the 200 test sinograms with a fixed seed (deterministic; every method sees the identical noisy input). RMS of the sinogram perturbation ≈ 0.012 (line-integral units), $\sim 1.8\%$ at the thickest ray. Supervised methods load their noiseless checkpoint and skip training; per-scene/classical methods re-fit on the noisy input. The FBP baseline is recomputed from the noisy sinogram.

S7 | Agent-behavior — extended account

This account follows the autoresearch paradigm of automated open-ended scientific discovery³⁵ realized through LLM code agents that resolve real software tasks^{36,37} and physics-informed agentic development in adjacent imaging fields.³⁸

Strengths. Fast paper→code; strong hyper-parameter optimization; strict fixed-budget discipline; scales evaluation across many parallel benchmarks; follows well-decomposed instructions.

Weaknesses. Does not invent new methods. Must be forced to recombine (converges early and pads if unsupervised — one breast sub-run replayed identical configs until an audit caught it). CT-image vision misses obvious artifacts, so numbers are the source of truth. Long tasks must be decomposed or even a 1M-token context is exhausted. Overfits to the task/metric (not only an agent trait).

Human meta-strategies (repeatable, same on both datasets). Persistence/breadth (keep asking for more directions); recombination (the idea to assemble pieces); redirection (do not mimic the other dataset); auditing (catch padding, enforce budget, fix provenance). The defensible claim: agent = tireless, mechanistically reasoning, self-modifying executor; human = strategist + auditor.

S8 | Additional figures (supplement)

- S-Fig 1: per-iteration val-vs-test hr trajectory (selection honesty).
- S-Fig 2: Mayo significance forest plot + solver×metric matrix.

- S-Fig 3: breast noiseless vs noisy example panels for all top-10 solvers.
- S-Fig 4: parameter-efficient climb (params-vs-hr Pareto), both datasets.
- S-Fig 5: effect-size (Cohen d_z) vs. raw Δ hr, illustrating the $n = 5$ vs $n = 200$ contrast.

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